

Null-Bridge Cosmology: A Recursive Pre-Bang Framework for Emergent Spacetime, Time, Mass, and Physical Constants

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****Status:**** Conceptual Foundations Paper (Work in Progress)

****Scope:**** Pre-bang ontology with emergence of time, space, mass, matter, gravity, fields, and constants from recursive null dynamics; compatible with GR, QM, QFT without post-inflation assumptions.

****Version:**** Unified Draft 1.0 (Integrating Null-Bridge, Null-Based, and Interaction-Limited Frameworks)

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This is an unreviewed AI generation from previous work in an attempt to unify and bolster the concept paper.

****Abstract****

We propose a minimal pre-bang framework where physical reality emerges from a non-integrable null prestate containing total potential yet forbidding differentiation. The null's logical instability necessitates a zero-dimensional relational construct (the "bridge" or $\text{Dim}[0]$), whose bounded, non-convergent recursion defines primitive time as an ordering parameter. Spacetime arises as projections of bridge states into effective configurations with multiple degrees of freedom, while mass emerges conditionally for persistence-enforcing geometries requiring sustained interaction with vacuum fluctuations. Gravity manifests as gradients of spatial inflow for persistent scaffolds, fields as transient bridges exporting phase corrections, and constants as computational boundaries from lattice geometry and recursion delays. This recursive model yields a tri-axial hexagonal lattice for emergent space, explains thermodynamic arrows via coherence loss, and provides testable predictions like coherence-dependent time dilation and nonlocal lensing, all without invoking temporal singularities, inflatons, or exotic particles.

****Highlights (for Reviewers)****

- Core Claim: Reality as self-resolving recursion converting null contradiction into structure; spacetime, matter, and constants as emergent coherence artifacts.
- Mechanism: Null paradox $\rightarrow$ extrusion of $\text{Dim}[0]$ recursion $\rightarrow$ quanta nodes $\rightarrow$ phase-locked loops $\rightarrow$ tri-planar hexagonal lattice (space) $\rightarrow$ persistent semi-closed loops (mass/matter).
- Gravity: Inflow gradients for persistence, not curvature postulate.
- Fields: Transient bridges for phase corrections; filaments as one-way decompression channels.
- Time: Primitive recursion index (real time) vs. emergent mass-time from record persistence.
- Constants: Derived from geometry/delays (e.g., c^* as max phase-handoff rate).
- Thermo: Heat as spin volatility; entropy as coherence loss.

1. Introduction and Core Hypothesis

We propose a formal framework describing the implicit infinite information in mathematics, extended to a pre-physical null prestate whose non-integrability drives recursive emergence. Unlike physical-instantiation claims, this model focuses on structural, relational, and combinatorial aspects of possibility space, serving as a tool for infinite pattern exploration without triviality or metaphysics.

Hypothesis: The universe emerges from a paradoxical null state ($\text{Dim}[-1]$), unstable due to containing all potentials including its negation. This instability extrudes a minimal 1D recursion conduit ($\text{Dim}[0]$ /bridge), iteratively resolving contradiction via self-reference. From this conduit's tick-dynamics arise quantized nodes, phase-locked loops, and a tri-axial hexagonal lattice perceived as 3D space. Matter corresponds to semi-closed recursive loops maintaining coherence by drawing spatial quanta inward (mass as persistence), gravity as macroscopic inflow signature, and fields as transient bridges carrying phase corrections. Physical constants are computational boundaries from lattice geometry and delays.

Dimensional Indexing:

- $\text{Dim}[-1]$: Null potential ($\{0D\}$).
- $\text{Dim}[0]$: 1D recursion conduit (bridge/substrate tick).
- $\text{Dim}[1-3]$: Emergent tri-axial coherence $\rightarrow$ spatial axes.

Time Distinction:

- Real Time: Invariant substrate tick of $\text{Dim}[0]$.
- Mass-Time: Emergent from record persistence in subsystems.

This framework is structurally compatible with General Relativity (GR), Quantum Mechanics (QM), and Quantum Field Theory (QFT), addressing initial conditions, time origin, mass gaps, and cosmic structure without singularities or exotic fields.

2. Foundational Assumptions and Definitions

****Definition 1 (Non-Integrable Null)****

Let $\mathcal{N}$ denote a pre-physical null prestate:

$\nexists \mu$ (σ -additive measure) such that $\mu(\mathcal{N})$ is finite and well-defined.

This implies $\mathcal{N}$ cannot be globally integrated or normalized, not equating to zero but forbidding closure. Interpretation: Null as logical instability containing $\infty \wedge \neg \infty$.

****Definition 2 (Opposed Subsets)****

Null admits decomposition into mutually opposed subsets: $\mathcal{N} \rightarrow \{A, \bar{A}\}$, not Boolean complements but unresolved oppositions failing joint totality.

****Definition 3 (Bridge State/ $\text{Dim}[0]$)****

Bridge $B \equiv \text{Rel}(A, \bar{A})$ as minimal relational construct, zero-dimensional, preceding spatial structure. Extrusion from null instability forms 1D recursion conduit.

****Definition 4 (Primitives and QuantaNodes)****

Overlaps of $\text{Dim}[0]$ paths stabilize as QuantaNodes: minimal phase-state sites carrying phase $(\theta, \Delta\theta)$, spin $(\Delta\theta_x, \Delta\theta_y, \Delta\theta_z)$, charge (persistent $\Delta\theta$ drift), sleep (low-volatility coherence).

Definition 5 (Recursive Update)

Deterministic operator $\mathcal{F}: \mathcal{B} \rightarrow \mathcal{B}$, $B_{n+1} = \mathcal{F}(B_n)$. Requirements: No fixed point, bounded orbit (e.g., rotations on S^1). Tick τ : Substrate update interval.

Definition 6 (Primitive Time)

Real time $t_R := n$ as recursion index; ordering parameter without metric or continuum.

Definition 7 (Realizability Boundary)

Distinction between formal existence and physically expressible patterns, bounded by laws and constraints.

3. Core Claims

- Every informational pattern corresponds to a numerical index in math space, deriving meaning through relational context.
- Transforms/decoders embedded in the index create closed, self-referential structure.
- Material systems (minds, AI) as constrained samplers of infinite backdrop.
- Patterns unrealizable in specific universes despite formal existence.
- Universe as self-resolving recursion: $\text{Null} \rightarrow \text{Dim}[0] \rightarrow \text{Nodes} \rightarrow \text{Loops} \rightarrow \text{Lattice} \rightarrow \text{Persistent scaffolds (matter)}$.

(Note: Meaningful patterns require transforms; complexity relates to $n+1$ indexing.)

4. Phase as the First Emergent Invariant

(Placed after recursive dynamics, before projection.)

The recursion admits no fixed point but is bounded, necessitating an invariant: phase ϕ .

4.1 Definition of Phase

Equivalence: $B_i \sim B_j \iff \exists k \in \mathbb{Z}, B_i = \mathcal{F}^k(B_j)$. $\phi(B_n) \in \mathbb{R}/2\pi\mathbb{Z}$: Ordering modulo recurrence; not spatial/temporal/energetic.

4.2 Necessity

Bounded non-convergence requires distinguishability; phase minimal sufficient invariant.

4.3 Topological Character

Phase space compact, continuous, boundaryless, oriented: Minimal manifold S^1 . Closure: $\oint d\phi = 2\pi$. Irrationality of ϕ reflects non-finite linear measure.

****4.4 Properties****

1. Projection invariance: $\phi(B_n) = \phi(\Sigma_n)$.
2. Additivity: $\phi_{n+m} = \phi_n + \phi_m \pmod{2\pi}$.
3. Metric independence.
4. Minimal increment $(\Delta\phi_{\min} > 0)$ for events.

****4.5 Relation to Physical Quantities****

Post-projection: $(\phi = S/\hbar)$, action (S) and $(\hbar)$ emergent.

****4.6 Role in Events****

Phase enables distinguishability; events as phase-induced observable changes.

5. Bounded Phase Variation and Emergent Propagation

Phase variation bounded: $(|\Delta\phi_n|_{S^1} \leq \Phi_{\max}) \in (0, \pi]$.

****5.1 Consistency****

Unbounded variation causes projection instability, no persistence, event trivialization.

****5.2 Emergent Adjacency****

Neighborhood: $(B_{n+1} \in \mathcal{N}_{\phi(B_n)} \text{ iff } |\phi_{n+1} - \phi_n|_{S^1} \leq \Phi_{\max})$. Pre-geometric locality.

****5.3 Propagation Under Projection****

Bounded variation implies finite observable changes: $(|\mathcal{O}(\Sigma_{n+1}) - \mathcal{O}(\Sigma_n)| \leq K_{\mathcal{O}})$. Precondition for coherent transport.

****5.4 Finite Signal Speed****

In metric spacetime: $(|\partial\phi / \partial x| < \infty)$, $(|\partial\phi / \partial t| < \infty)$. Via $(\phi = S/\hbar)$: Finite max speed emergent from $(\Phi_{\max})$.

****5.5 Uniqueness of Adjacency****

Shortest arc distance on (S^1) : $(d_{S^1}(\phi_1, \phi_2) = \min_k |\phi_1 - \phi_2 + 2\pi k|)$. Minimal criterion for bounded recursion.

6. Emergent Decomposition and Projection Necessity

Recursion remains global without internal degrees; projection expands representation for substructure.

****6.1 Requirement****

Needs: Distinguishability, adjacency, persistence. Single (ϕ_n) insufficient.

****6.2 Definition****

$(\mathcal{P}: \mathcal{B} \rightarrow \mathcal{S})$, $(\Sigma_n = \mathcal{P}(B_n))$. Conditions: Nontriviality, phase consistency, continuity w.r.t. phase.

****6.3 Decomposition****

$\Sigma_n \equiv (\sigma_{n^{\{1\}}}, \dots, \sigma_{n^{\{m\}}})$, $m > 1$. Prerequisite for locality/propagation.

****6.4 Implications****

Enables events as observable changes, locality as phase constraints, memory as correlations.

7. Projection and Emergent Spacetime

Projection from bridge to configuration space with degrees.

****7.1 Operator Definition****

$\mathcal{P}: \mathcal{B} \rightarrow \mathcal{S}$, many-to-one coarse-graining.

****7.2 Requirements****

(i) Phase preservation. (ii) Bounded variation compatibility. (iii) Decomposition support. (iv) No presupposed geometry.

****7.3 Spacetime as Effective****

Emerges from regular families Σ_n ; coarse-graining where bounded variation $\rightarrow$ bounded changes interprets as space/time. Geometry emergent descriptive layer.

****7.4 Local Update and Invariant Throughput****

Admissible transitions phase-consistent; influence mediated by recursion, not instantaneous.

Proposition: Minimal $\Delta n_{\min} \geq 1$; influence only if $\Delta n \geq \Delta n_{\min}$.
Emergent: Finite throughput $c \equiv \ell / (\Delta n_{\min} \tau)$. Invariant w.r.t. recursion time; varies with mass-time density (dilation).

****7.5 Empirical Anchoring****

Determines existence/role of scales/bounds; numerical values via anchoring to units.

****7.6 From Null to Lattice****

1. Contradiction: $\{0D\} = \infty \wedge \neg \infty$.
2. Extrusion: Self-reference $\rightarrow \text{Dim}[0]$.
3. Nodes: Path overlaps stabilize QuantaNodes.
4. Looping: 3-node minimal loops persist.
5. Lattice: Symmetry/throughput select tri-axial hex tiling (6 in-plane neighbors, offset stacking).
Why hex? Maximizes phase equalization, supports slippage/flow, densest packing with shear.

8. Events and Minimal Change

Events definable post-projection.

****8.1 Observables****

$\backslash(\mathcal{O}: \mathcal{S} \rightarrow \mathbb{R})$; on projections only.

****8.2 Event Definition****

Event at $\backslash(n)$ if $\backslash(\exists \mathcal{O}, \mathcal{O}(\Sigma_{n+1}) \neq \mathcal{O}(\Sigma_n))$.
Minimal distinguishable change.

****8.3 Minimality****

Necessary post-projection; sufficient via single change; non-redundant.

****8.4 Relation to Phase/Adjacency****

Constrained by bounded variation; no causality/asymmetry yet.

****8.5 Pre-Temporal****

Events precede experienced time; sequences gain arrow via memory.

9. Energy as Unresolved Differential

Energy from persistent null asymmetry.

****9.1 Tension****

$\backslash(\mathcal{Q}(B_n) = D(A, \bar{A}; B_n) \geq 0)$; vanishes on resolution (not globally).

****9.2 Projection to Energy****

Distributed as density; configuration property, not trajectory.

****9.3 Minimality****

Necessary for unresolved opposition; sufficient scalar; non-redundant.

****9.4 Later Forms****

Decomposes to kinetic/potential/field post-spacetime.

10. Capacity and Threshold Dynamics

Prevents divergence.

****10.1 Finite Capacity****

For subregion $\backslash(\mathcal{R} \subset \Sigma_n)$, $\backslash(\mathcal{Q}_{\mathcal{R}} \leq C_{\mathcal{R}} < \infty)$.

****10.2 Threshold****

If $\backslash(\mathcal{Q}_{\mathcal{R}} \geq \mathcal{Q}_{\text{crit}}(\mathcal{R}))$, reconfigure to admissible $\backslash(\Sigma_{n+1})$.

****10.3 Non-Singularity****

Enforces finite admissibility without divergence.

****10.4 Later Phenomena****

Phase transitions, particle creation as effective interpretations.

11. Record Persistence and Mass-Time

Arrow from asymmetry.

****11.1 Retention****

Subsystem record $\set(R_n \in \mathcal{R})$; update $\set(R_{n+1} = \mathcal{G}(R_n, \Sigma_{n+1}))$, monotonic $\set(R_{n+1} \supseteq R_n)$. Local, contingent.

****11.2 Temporal Arrow****

Partial order $\set(n_1 \prec_M n_2 \text{ iff } R_{n_2} \supset R_{n_1})$. Mass-time $\set(t_M)$; directional where defined.

****11.3 Mass Relation****

Persistence requires stability/tension; mass inseparable from records. Local emergence.

****11.4 Distinction****

$\set(t_M)$ emergent/subsystem-dependent; causality where $\set(t_M)$ and interactions exist.

****11.5 Classes of Behavior****

(A) Trivial: No closure/primitive.

(B) Massless: Closure, no history; local interactions (e.g., photons).

(C) Massive: Persistence-enforcing; interaction cycles $\rightarrow$ inertia/gap.

****11.6 Mass Gap Origin****

No intermediate regimes; cost irreducible once chained.

****11.7 Speed Bound****

Finite coordination for chains; asymptotic velocity limit for massive.

12. Mass, Gravity, and Flow Engines

Mass: Persistence cost of semi-closed scaffolds importing quanta.

Gravity: Inflow gradient; bookkeeping of recursive demand.

Flow Engines: Route quanta, expel excess via filaments (one-way decompression).

13. Fields, Memory, Nonlocality

Fields: Transient $\text{Dim}[0]$ bridges exporting corrections.

Memory: Phase templates for synchronized updates.

Entanglement: Shared history to templates; parallel resolutions.

14. Constants as Boundaries

c: Max phase-handoff $\set(\ell_P / \tau)$.

h: Min loop action (6-node ring over 6τ).

G: Inflow efficiency vs. coherence.

$(\alpha, \epsilon_0, \mu_0)$: Lattice symmetry/delays.

Goal: Derive from (ℓ_P, τ) , symmetry.

15. Thermodynamics

Heat: Spin volatility/phase noise.

Entropy: Coherence loss; unresolved recursion spread.

Gradients: Flow to volatile zones.

Fate: Low-throughput plateau; voids as sinks/seeds.

16. Expansion and Dark Energy

Expansion: Low-density inflation from quanta draw.

Dark Energy: Residual tension in voids (bookkeeping).

Feedback: Limits acceleration; asymptote.

17. Implications for Information Theory/Computation

Pattern recovery: Surfacing from index.

AI: Partial IEoll exploration via token prediction; prompt as transform selection.

Complexity: Local relational expressions.

Cross-Links: Holographic principle (boundary info), evolution as sampler.

18. Correspondence with Physics

****GR:**** Geometry from projection; singularities as reconfigurations.

****QM:**** Time as parameter; measurement as records; decoherence via proliferation.

****QFT:**** Vacuum as bridge oscillation; excitations as tension projections; zero-point as unresolved without records.

Mass Gap: Irreducible persistence cost (Yang-Mills mapping).

Scope: Structural, not derivational; emergent assumptions.

19. Predictions and Falsifiability

1. Coherence-dependent dilation (temperature-independent).

2. Nonlocal lensing (filament exits, no baryons).

3. Void-biased seeding.

4. CMB anisotropies from templates.

5. Lab phase transport anomalies in hex metamaterials.

6. One-way channeling in analogs.

Falsified by: No field dynamics from projections; threshold mismatch to phases; record failure for thermo arrow; contradictory cosmology.

20. Conclusion

This unified framework conceptualizes infinite possibility as relational, embedded in math space, with physical emergence from null recursion. Distinguishing formal from realized, it explores emergence, pattern, creativity from infinite backdrop without triviality/inflation.

****Acknowledgments:**** [To be added]

****References:****

- Aristotle on potential infinity.
- Cantor on transfinite numbers.
- Peano arithmetic (successor).
- Turing on computability.
- Chaitin on algorithmic info.
- Kolmogorov complexity.
- Tegmark Mathematical Universe.

(Additional: Stanford/IEP on Infinity; IntuitionMachine on conceptual shortcuts; arXiv on emergent time/cosmology; recursive hierarchical models.)

****Appendix A: Minimal Discrete Model****

Substrate: Lattice nodes with state/history.

Primitives: Closed dependencies.

Vacuum: Baseline return unless retained.

Interactions: Propagation, retention, inhibition.

Cost: History count; gap below/above threshold.

Bound: Finite resolution rate.

Massless: History-free propagation.

Demonstrates: Closure $\neq$ mass; deficits force ingress; persistence $\rightarrow$ inertia/gap/bound.

****Mathematical Appendix****

A.1 Null Extrusion: Contradiction compels recursion domain.

A.2 Node/Loop: 3-node min; hex from throughput/min variance $\backslash(E = \sum |\theta_i - \theta_j|^2)$.

A.3 Constants: Indicative relations; derive from tick arithmetic.

A.4 Action: $\backslash(S = \sum \Delta\tau (E_{n^{\text{real}}} - E_{n^{\text{con}}}))$; macro yields $\backslash(L = T - V)$.

****Glossary****

- Bridge/Dim[0]: Recursion conduit.
- QuantaNode: Phase/spin/charge/sleep site.
- Loop: Persistent cycle (identity).
- Semi-Closed Flow: Quanta-drawing scaffold (mass).
- Filament: One-way channel.
- Real vs. Mass-Time: Tick vs. throughput.

****Change Log****

- Unified null-bridge with recursive lattice/mass mechanisms.
- Integrated phase/record for emergent time/mass.
- Added predictions/roadmap for falsifiability.
- Consolidated thermo/expansion/dark energy.

